

KARUNADA-KALA

*AI-Powered Cultural Ecosystem Platform
for Karnataka Heritage*

PRODUCT REQUIREMENTS DOCUMENT

Developed By	Eshaan Mayekar
Organization	MindMatrix Internship Program
Project Type	Final Internship Project
Platform	Android Application
Version	1.0 · 2026

Preserving Karnataka's Living Heritage Through Modern Technology

SECTION 01

Project Overview

Karunada-Kala is an AI-powered Android application developed to preserve, promote, and digitally showcase the rich cultural heritage of Karnataka. Traditional art forms such as Yakshagana, Dollu Kunittha, Bidri Art, Kinnala Toys, Lambani Craft, and many other regional cultural identities are gradually losing visibility among younger generations and modern audiences.

The application acts as a digital cultural ecosystem where users can discover Karnataka's heritage through interactive exploration, maps, events, workshops, artisan profiles, and AI-powered cultural assistance. The platform combines modern Android technologies, cloud services, and Generative AI to create an immersive and engaging cultural experience.

Technology Stack

Kotlin	Jetpack Compose	Firebase Firestore	Firebase Auth
Firebase Storage	Google Gemini AI	Google Maps API	MVVM Architecture
Android Studio	Git & GitHub	Material Design 3	Shared Transitions

SECTION 02

Problem Statement

Traditional Karnataka art forms and cultural practices are struggling to reach modern audiences due to lack of digital visibility, limited awareness, and reduced accessibility. Many users are unaware of authentic artisans, local workshops, cultural festivals, and heritage learning opportunities available across Karnataka.

The platform addresses the following gaps:

- Preserve Karnataka's cultural identity in the digital space
- Promote traditional art forms to younger and global audiences
- Connect users directly with artisans and local workshops
- Increase awareness about cultural festivals and heritage events
- Provide modern, accessible entry points to traditional knowledge
- Encourage cultural tourism and community participation

Karunada-Kala solves this problem by creating an AI-powered cultural discovery and preservation platform that bridges the gap between Karnataka's living heritage and its modern digital audience.

SECTION 03

Objectives

01. Create a centralized digital platform for Karnataka cultural heritage
02. Preserve traditional art forms using modern Android and AI technologies
03. Provide users with rich cultural exploration and discovery features
04. Integrate AI-powered assistance, insights, and personalized recommendations
05. Connect users with artisans, performers, and workshop facilitators
06. Provide event discovery and workshop registration functionality
07. Develop a visually engaging, Karnataka-themed Android application
08. Implement real-time cloud database integration using Firebase
09. Improve accessibility and cultural awareness among users

SECTION 04

Scope of the Project

The application is designed as a scalable MVP (Minimum Viable Product) that can later expand into a complete cultural startup platform. The current scope includes:

• Cultural art exploration	• Artist and artisan discovery
• Cultural event browsing	• Workshop registration
• AI-powered recommendations	• Interactive Google Maps integration
• Community engagement system	• User profile and milestone tracking
• Firebase cloud synchronization	• Responsive Android UI/UX

Target Users

✓ Students and learners	✓ Tourists and travelers
✓ Art and culture enthusiasts	✓ Researchers and historians
✓ Local artisans and performers	✓ Cultural organizations
✓ Workshop participants	✓ Karnataka heritage communities

SECTION 05

Functional Requirements

7.1 User Authentication

- User login and signup functionality
- Firebase Authentication integration
- User profile management and session handling

7.2 Explore Section

- Browse cultural art forms with rich visual cards
- Search functionality and category filtering
- AI-based cultural recommendations
- Interactive cultural content cards

7.3 Cultural Maps

- Google Maps integration for Karnataka
- Display artisan, workshop, and performance locations
- Distinct cultural map markers by category

7.4 Events Section

- Display upcoming festivals and cultural events
- Event registration and details page
- Cultural event recommendations

7.5 Artist Profiles

- Artist biography, cultural lineage, experience
- Tap-to-call and WhatsApp contact support
- Workshop details and availability

7.6 Community Section

- Share cultural moments via community feed
- Image sharing and cultural storytelling

7.7 AI Features

- Google Gemini AI integration

- AI-generated cultural insights and summaries
- Personalized cultural recommendations

7.8 Journey Dashboard

- Cultural milestone tracking and achievement badges
- Participation statistics and personalized progress

SECTION 06

System Architecture — MVVM

The application follows the MVVM (Model-View-ViewModel) architecture pattern, which is the industry-standard approach for modern Android development with Jetpack Compose. This architecture ensures clean separation of concerns, testability, and scalability.

VIEW LAYER

- Jetpack Compose UI Screens
- Navigation Components
- Shared Element Transitions
- Interactive UI Elements

VIEWMODEL LAYER

- State Management using StateFlow
- Business Logic and Data Processing
- API Handling and Orchestration
- Error Handling and Loading States

MODEL / REPOSITORY LAYER

- Firebase Firestore Data Models
- Repository Pattern Abstraction
- Remote & Local Data Mapping

EXTERNAL SERVICES

- Firebase Auth, Firestore, Storage, FCM
- Google Maps API
- Google Gemini AI API

SECTION 07

UI/UX Design Approach

The UI/UX design of Karunada-Kala was inspired by Karnataka's cultural aesthetics while maintaining a modern and premium mobile application experience. The design balances cultural identity with clean, contemporary Android design principles.

Design Principles

- Clean and minimal layouts with cultural visual identity
- Karnataka-inspired color palette — Deep Red, Cream White, Heritage Gold, Forest Green
- Smooth navigation flow with shared element animations
- Soft glassmorphism effects and rounded modern components
- Responsive design across various Android screen sizes
- Interactive cultural exploration with immersive experience

Theme Color Palette

Deep Karnataka Red	#8B1A1A	Primary brand, headings, CTAs
Heritage Gold	#B8860B	Ornaments, highlights, borders
Forest Green	#2D5A27	Subheadings, success states
Cream White	#FAF6EE	Background, card surfaces

SECTION 08

Database Design — Firebase Firestore

Firebase Firestore was used as the cloud database solution, providing real-time data synchronization across all application screens. The database follows a collection-document model optimized for the application's query patterns.

arts	Stores cultural art forms — title, description, category, media, AI insights
artists	Artisan and performer profiles — bio, lineage, location, contact, achievements
events	Cultural events and festivals — name, date, location, registration details
workshops	Workshop listings — facilitator, schedule, capacity, registration data
registrations	User registration records for events and workshops
chronicles	Community-shared cultural posts — image, story, user reference

SECTION 09

Features Implemented

Core Features

- Onboarding screens with Karnataka cultural theme
- Explore dashboard with cultural discovery cards
- Cultural search and category filtering system
- Event browsing and workshop registration
- Google Maps — Artisan Map with cultural markers
- Firebase Firestore database integration
- Artist profiles with contact and cultural lineage
- AI-powered suggestions using Gemini API
- Community section with cultural feed and sharing
- Journey dashboard with milestone tracking
- Achievement badges and participation statistics
- Personalized user profile management

Advanced & Technical Features

- Dynamic UI transitions and shared element animations
- Cultural-themed color system — Karnataka heritage palette
- Real-time cloud synchronization using Firebase Firestore
- AI-powered recommendations via Google Gemini AI
- Modern bottom navigation with animated indicators
- Kotlin Coroutines and StateFlow for async operations
- MVVM architecture for clean code separation
- Firebase Cloud Messaging for push notifications

SECTION 10

Challenges Faced & Solutions

■ Understanding MVVM Architecture & StateFlow

- ✓ **Solution:** Studied official Android documentation, applied pattern consistently across all screens, and used structured ViewModels for state management.

■ Implementing Responsive UI/UX Screens

- ✓ **Solution:** Used Jetpack Compose's adaptive layout system, tested on multiple emulator screen sizes, and iterated on component spacing.

■ Firebase Firestore Integration

- ✓ **Solution:** Followed official Firebase documentation, implemented repository pattern to abstract data operations, and structured queries for efficiency.

■ Managing Real-Time Cloud Updates

- ✓ **Solution:** Used Kotlin Flows and Firestore snapshot listeners to propagate real-time updates through the ViewModel to the UI layer.

■ Google Maps API Integration

- ✓ **Solution:** Implemented Maps Compose library, handled location permissions properly, and designed custom cultural map markers.

■ Integrating Generative AI Services

- ✓ **Solution:** Used Gemini AI SDK with proper prompt engineering to generate culturally relevant content and summaries within the app context.

■ Optimizing Application Performance

- ✓ **Solution:** Applied lazy loading for image content, optimized Firestore queries with proper indexing, and minimized unnecessary recompositions in Compose.

■ Debugging Android Crashes

- ✓ **Solution:** Used Android Logcat systematically, applied structured error handling with try-catch and proper null safety in Kotlin.

SECTION 11

Testing & Debugging

The application was tested using Android Emulator and physical Android devices to ensure stable performance across different screen sizes and Android versions.

UI Testing	Verified composable rendering, layout responsiveness, and visual consistency
Navigation Testing	Validated all screen transitions, back stack behavior, and deep links
Firebase Sync Testing	Confirmed real-time Firestore reads/writes and update propagation
API Response Testing	Tested Gemini AI and Maps API responses with edge case handling
Performance Testing	Monitored screen load times, image rendering, and memory usage
Error Handling	Verified graceful handling of network errors, null responses, auth failures

SECTION 12

Outcomes & Achievements

The project successfully achieved all primary objectives set at the beginning of the internship program. Karunada-Kala stands as a fully functional, AI-powered Android application demonstrating real-world product development capability.

Functional Android Application	Complete multi-screen app running on Android emulator and devices
AI-Powered Cultural Discovery	Gemini AI integrated for insights, recommendations, and content
Firebase Cloud Backend	Firestore, Auth, Storage, and FCM fully integrated and operational
Google Maps Integration	Artisan Map with custom markers covering Karnataka cultural locations
Modern UI/UX System	Karnataka-themed design with responsive layouts and smooth animations
MVVM Architecture	Clean, scalable, and maintainable code structure throughout
Real-Time Synchronization	Live data updates across all sections of the application
Cultural Impact	Digital platform for Karnataka heritage preservation and discovery

SECTION 13

Future Enhancements

Karunada-Kala is designed as a scalable foundation. The following enhancements are planned to evolve the platform into a complete cultural startup product:

Multi-Language Support	Kannada, Hindi, and regional language support for wider reach
AI Cultural Chatbot	Dedicated Gemini AI-powered cultural assistant for user queries
Voice Assistant Integration	Voice-based navigation and cultural storytelling features
AR/VR Cultural Experiences	Augmented reality for immersive heritage site exploration
Digital Artisan Marketplace	Platform for artisans to sell traditional crafts directly
Real-Time Workshop Booking	Live booking system with payment gateway integration
Offline-First Architecture	Room Database for complete offline content access
Social Media Integration	Share cultural discoveries directly to social platforms
Personalized AI Engine	Advanced recommendation system based on user behavior

SECTION 14

Tools & Technologies Reference

Programming Language	Kotlin
UI Framework	Jetpack Compose + Material Design 3
IDE	Android Studio
Database	Firebase Firestore (NoSQL, Real-time)
Authentication	Firebase Authentication
Storage	Firebase Storage
Push Notifications	Firebase Cloud Messaging (FCM)
AI Integration	Google Gemini AI API
Maps	Google Maps Compose API
Architecture	MVVM (Model-View-ViewModel)
Async Operations	Kotlin Coroutines & StateFlow
Version Control	Git & GitHub
Design Tools	Google Stitch, Figma Concepts

SECTION 15

Conclusion

Karunada-Kala was successfully developed as an AI-powered cultural ecosystem platform focused on preserving and promoting Karnataka's traditional heritage through modern Android technologies. The project combined Firebase cloud services, Google Maps integration, Generative AI features, and modern UI/UX principles to create an engaging and interactive user experience.

The application demonstrated practical implementation of Android development concepts, cloud database integration, AI-powered systems, and responsive mobile UI/UX design. The project provided valuable exposure to real-world software development workflows, problem-solving techniques, and modern technology integration.

Overall, Karunada-Kala successfully achieved its objectives and serves as a scalable foundation for future cultural preservation and digital heritage platforms. The project not only demonstrates technical capability but also reflects a meaningful contribution toward preserving Karnataka's living cultural identity for future generations.

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Appendix

Application Screens

01	Splash Screen	Karnataka-themed branded entry experience
02	Onboarding Screen	Cultural introduction with value propositions
03	Explore Dashboard	Main cultural discovery hub with AI recommendations
04	Cultural Maps Screen	Google Maps with artisan and cultural location markers
05	Events Screen	Cultural festivals and event listing with registration
06	Artist Detail Screen	Full artisan profile with contact and cultural history
07	Workshop Registration	Workshop booking and participation system
08	Community Screen	Cultural community feed and story sharing
09	Journey Dashboard	Milestone tracking and achievement badges
10	User Profile Screen	Personalized user account and preferences

Deliverables

- Android Application (APK) — Functional application ready for device installation
- Source Code Repository — Organized GitHub repository with complete codebase
- Internship Report — Detailed documentation of activities and learnings
- Project PRD — This Product Requirements Document
- Firebase Backend — Configured Firestore database with sample cultural data
- UI/UX Design System — Karnataka-themed color and component system

References

- Android Developer Documentation — developer.android.com
- Firebase Documentation — firebase.google.com/docs
- Google Maps API Documentation — developers.google.com/maps
- Google AI Studio & Gemini Documentation — ai.google.dev

- [Jetpack Compose Documentation](https://developer.android.com/jetpack/compose) — developer.android.com/jetpack/compose
- [Material Design 3 Guidelines](https://m3.material.io) — m3.material.io
- [MindMatrix Internship Resources](#)

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